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# SKIN LESION CLASSIFICATION USING SWIN TRANSFORMER ON HAM10000 DATASET

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**The domain of the Project:**  
Deep Learning & Medical Image Classification

**COURSE NAME**  
G36 Python

**Team Mentors (and their designation):**  
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**Team Members:**  
Mr. Y. Dileep Kumar

**Period of the project:**  
20 Days

**March 2026    to    April 2026**

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## Declaration

The project titled “*Skin Lesion Classification using Swin Transformer on HAM10000 Dataset*” has been mentored by Bhargavesh Dakka sir, organised by SURE Trust, from March 2026 to April 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

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## Executive Summary

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### Project Objectives

The primary goal of this project was to build an end-to-end deep learning pipeline that classifies dermoscopic skin lesion images from the HAM10000 dataset into seven diagnostic categories using the Swin Transformer architecture. By combining transfer learning with fine-tuning on imbalanced medical imagery, the project aimed to deliver:

- **Educational Deep Learning Pipeline:** Demonstrate how a Vision Transformer model can be adapted for a real-world medical classification task with a structured, reproducible training workflow.
- **Clinical-Grade Image Analysis:** Apply class-weighted loss functions and data augmentation to handle severe class imbalance and produce a reliable skin lesion classifier.

### Methods & Approach

To achieve high performance on the imbalanced HAM10000 dataset, the system was built using PyTorch and the `timm` library without heavy custom frameworks. The core technical approach included:

- **Two-Phase Transfer Learning:** Phase 1 freezes the Swin Transformer backbone and trains only the classification head for 5 epochs at `lr=1e-3`. Phase 2 unfreezes all layers for full fine-tuning over 15 epochs using AdamW with Cosine Annealing.
- **Class-Weighted Loss:** Inverse-frequency class weights were computed to address the extreme imbalance in HAM10000 (*nv* class alone comprises  $\sim 67\%$  of samples), and passed into `CrossEntropyLoss`.
- **Robust Data Augmentation:** Training images were subjected to random flips, rotations ( $\pm 15^\circ$ ), and colour jitter to improve generalisation across diverse lesion appearances.

### Key Findings & Architectural Successes

- **State-of-the-Art Backbone:** Using `swin_base_patch4_window7_224` (pretrained on ImageNet-22k) provided powerful hierarchical representations out-of-the-box.
- **Stable Training Loop:** The best validation weights were checkpointed automatically during Phase 2, ensuring the final model corresponds to peak generalisation rather than last-epoch overfitting.
- **End-to-End Inference:** A fully self-contained prediction script allows a new dermoscopic image to be uploaded and classified with a softmax confidence score for all seven lesion types.

## Strategic Recommendations (Future Scaling)

| # | Initiative                | Description                                                                                                                                     |
|---|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Pyodide / ONNX Deployment | Export the trained model to ONNX and serve it via a lightweight FastAPI endpoint or a browser-based ONNX Runtime, eliminating GPU server costs. |
| 2 | Grad-CAM Explainability   | Integrate Gradient-weighted Class Activation Maps to highlight the exact lesion region driving each prediction, increasing clinical trust.      |
| 3 | Dataset Expansion         | Incorporate ISIC 2019/2020 datasets and apply MixUp or CutMix augmentation to further reduce class imbalance and improve minority-class recall. |

## Conclusion & Business Impact

This project successfully positions itself at the intersection of Computer Vision and Medical AI. By adopting a two-phase fine-tuning strategy on a pretrained Swin Transformer, the pipeline delivers a clinically motivated, reproducible skin lesion classifier with strong handling of class imbalance and a clear pathway to production deployment.

# Introduction

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## Background and Context of the Project

Skin cancer is among the most common forms of cancer globally, yet early detection dramatically improves patient outcomes. Dermoscopy—a non-invasive imaging technique—produces high-resolution images of skin lesions that carry rich diagnostic information. However, manual diagnosis by dermatologists is time-consuming, subject to inter-observer variability, and inaccessible in resource-limited settings. The HAM10000 (Human Against Machine with 10 000 training images) dataset, curated by the ISIC Archive, provides a benchmark for automated multi-class lesion classification across seven categories. Simultaneously, Vision Transformers—specifically the Swin Transformer—have emerged as the gold-standard backbone for medical image analysis, outperforming CNN-based architectures on fine-grained visual recognition tasks.

## Problem Statement and Goals

**Problem Statement:** Automated skin lesion classification faces two critical challenges: (1) severe class imbalance—the Melanocytic Nevi (*nv*) class represents  $\sim 67\%$  of HAM10000 samples while rare classes such as Dermatofibroma (*df*) and Vascular Lesions (*vasc*) are severely under-represented; and (2) the need for a high-capacity feature extractor that captures both local texture and global structure in dermoscopic images.

**Project Goals:** The primary objective is to engineer a full PyTorch training pipeline capable of:

- **Transfer Learning:** Fine-tune a pretrained Swin Transformer (`swin_base_patch4_window7_224`) on HAM10000 images using a two-phase strategy.
- **Class Imbalance Handling:** Compute and apply inverse-frequency class weights within `CrossEntropyLoss` to prevent majority-class collapse.
- **Rigorous Evaluation:** Generate a confusion matrix and per-class classification report (precision, recall, F1) on the held-out validation split.
- **Inference Pipeline:** Deliver a standalone prediction script that accepts any new dermoscopic image and returns the predicted lesion category with confidence scores.

## Scope and Limitations

### Scope of the Project:

- The pipeline is designed for the HAM10000 benchmark dataset with seven lesion classes.
- Training is conducted on GPU (CUDA) via Kaggle Notebooks, using the `timm` and `torchvision` libraries.
- The architecture uses `swin_base_patch4_window7_224` with ImageNet-22k pretrained weights.
- Evaluation is performed on a stratified 80/20 train-validation split.

### Limitations of the Project:

- **External Libraries:** The pipeline requires GPU execution; CPU-only inference is significantly slower.
- **Dataset Scope:** The model is trained and evaluated exclusively on HAM10000; generalisation to clinical imaging systems or other datasets has not been validated.
- **Class Imbalance:** Despite class-weighted loss, the rarest classes (*df*, *vasc*) may still show lower recall due to absolute sample scarcity.
- **Explainability:** The current implementation does not include saliency maps or Grad-CAM; the model functions as a black-box classifier.

### Innovation Component

The pipeline introduces several engineering decisions that elevate it above a standard fine-tuning tutorial:

- **Two-Phase Fine-Tuning Strategy:** Rather than fine-tuning all layers from the start (which can destroy pretrained representations on small datasets), Phase 1 warms up only the classification head before Phase 2 gradually unlocks the entire backbone. This mirrors the LLM instruction-tuning paradigm applied to vision.
- **Dynamic Class-Weight Computation:** Class weights are computed at runtime from the actual training split, making the pipeline robust to any resampling strategy without manual recalibration.
- **Cosine Annealing Scheduler:** The AdamW + CosineAnnealingLR combination in Phase 2 prevents sharp loss spikes during fine-tuning and achieves smoother convergence compared to step-decay schedules.
- **Best-Weight Checkpointing:** Model weights are saved only when validation accuracy improves, ensuring the deployed model corresponds to the best generalisation point rather than the final training epoch.

# Project Objectives

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## Core Objectives and Goals

The overarching goal of this project is to demystify medical image classification by building a rigorous, reproducible deep learning pipeline on a real clinical benchmark. To achieve this, the project targeted four specific technical and educational goals:

- **Clinical Transparency:** To produce a classifier that reports per-class precision, recall, and F1 scores, giving a medically meaningful performance profile beyond top-1 accuracy.
- **Imbalance-Aware Training:** To engineer a training loop that dynamically counteracts dataset imbalance using inverse-frequency class weights, preventing naive majority-class overfitting.
- **Transfer Learning Efficiency:** To demonstrate that a two-phase fine-tuning strategy (head-only → full model) outperforms naive end-to-end training on a limited medical dataset.
- **Production-Ready Inference:** To deliver a standalone, self-contained prediction script that can classify any new dermoscopic image without requiring re-loading the full training environment.

## Expected Outcomes

- **A Trained Swin Transformer Model:** A `.pth` checkpoint achieving competitive validation accuracy across all seven HAM10000 lesion categories.
- **Detailed Evaluation Report:** A per-class classification report and confusion matrix providing clinical insight into model strengths and weaknesses.
- **Reproducible Training Pipeline:** A structured notebook that can be re-executed on Kaggle with a single run to reproduce all results.
- **Plug-and-Play Inference:** An inference cell that accepts any uploaded image and returns the predicted class with full softmax probability distribution.

## Key Deliverables

The project culminates in the delivery of four core modules:

- **Data Preparation Module (Cell 0–1):** Kaggle dataset ingestion, metadata parsing, image organisation by class, and stratified 80/20 train-validation split.
- **DataLoader & Augmentation Module (Cell 2–3):** `torchvision` ImageFolder pipelines with training augmentations (flip, rotation, colour jitter) and class-weight computation for imbalanced `CrossEntropyLoss`.
- **Model Training Module (Cell 4–6):** Two-phase fine-tuning of `swin_base_patch4_window7_224`: head-only warm-up (5 epochs) followed by full fine-tuning with AdamW + CosineAnnealingLR (15 epochs) and best-weight checkpointing.



- **Evaluation & Inference Module (Cell 7–9):** Confusion matrix, per-class classification report, model persistence (.pth), and a standalone image prediction pipeline with confidence scores.

## Methodology and Results

### Methods and Technology Used

The pipeline was developed using a structured, GPU-accelerated PyTorch workflow on Kaggle Notebooks. The core methods and technologies include:

- **Model Architecture:** Swin Transformer Base (`swin_base_patch4_window7_224`) loaded via the `timm` library with ImageNet-22k pretrained weights and a 7-class output head.
- **Training Strategy:** Two-phase fine-tuning — Phase 1 trains only the classification head (Adam, `lr=1e-3`, 5 epochs) to establish a strong linear probe; Phase 2 unfreezes the full backbone (AdamW, `lr=1e-4`, 15 epochs, CosineAnnealingLR).
- **Loss Function:** `CrossEntropyLoss` with dynamically computed inverse-frequency class weights to address the severe class imbalance in HAM10000.
- **Data Augmentation:** `RandomHorizontalFlip`, `RandomRotation(15)`, `ColorJitter` (`brightness=0.2`, `contrast=0.2`), followed by ImageNet normalisation (`mean=[0.485, 0.456, 0.406]`, `std=[0.229, 0.224, 0.225]`).
- **Evaluation:** Scikit-learn `classification_report` (precision, recall, F1 per class) and `confusion_matrix` on the held-out validation split.

### Tools and Software Used

| Tool / Library   | Version   | Purpose                                                                                            |
|------------------|-----------|----------------------------------------------------------------------------------------------------|
| PyTorch          | 2.x       | Core deep learning framework — model, loss, optimiser, dataloader                                  |
| timm             | 0.9+      | Model zoo providing pretrained Swin Transformer weights                                            |
| torchvision      | 0.17+     | ImageFolder dataset, transforms, and normalisation                                                 |
| scikit-learn     | 1.x       | <code>train_test_split</code> , <code>classification_report</code> , <code>confusion_matrix</code> |
| kagglehub        | latest    | Dataset ingestion from Kaggle Hub API                                                              |
| CUDA             | 11.x/12.x | GPU acceleration for model training and inference                                                  |
| Kaggle Notebooks | —         | Execution environment with free P100 / T4 GPU                                                      |

### Data Collection Approach

The HAM10000 dataset was ingested directly from the Kaggle Hub using the `kagglehub` library, which provides a Pandas-native adapter for loading metadata CSVs. Images are distributed across two separate archive folders (`HAM10000_images_part_1` and `part_2`); the pipeline resolves each image’s location by checking both directories and copies the file into a class-organised directory structure under `/kaggle/working/HAM10000_split`. A stratified 80/20 split (`sklearn.train_test_split`, `random_state=42`) is then applied per class to maintain the original class distribution in both training and validation sets.

## Project Architecture

The pipeline is structured into five sequential stages:

| Stage            | Module   | Description                                                                                                                                                                               |
|------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 — Ingestion    | Cell 0   | Load metadata CSV via <code>kagglehub</code> , map image IDs to file paths, copy images into class subdirectories.                                                                        |
| 2 — Splitting    | Cell 1   | Apply per-class 80/20 stratified split; move images into <code>train/</code> and <code>val/</code> directory trees.                                                                       |
| 3 — Loading      | Cell 2–3 | Construct <code>torchvision</code> <code>ImageFolder</code> datasets with augmentation transforms; compute inverse-frequency class weights.                                               |
| 4 — Training     | Cell 4–6 | Build Swin Transformer with <code>timm</code> ; Phase 1 head-only warm-up (Adam, 5 epochs); Phase 2 full fine-tuning with AdamW + CosineAnnealingLR (15 epochs) + best-weight checkpoint. |
| 5 — Eval & Infer | Cell 7–9 | Generate confusion matrix and per-class report; save model <code>state_dict</code> ; run single-image inference with softmax confidence output.                                           |

## Key Code Snippets

### Model Initialisation:

```
model = timm.create_model(
    'swin_base_patch4_window7_224',
    pretrained=True,
    num_classes=7,           # HAM10000 classes
)
model = model.cuda()
```

### Class-Weighted Loss:

```
class_weights = [sum(class_counts) / c for c in class_counts]
class_weights = torch.tensor(class_weights, dtype=torch.float).cuda()
criterion = nn.CrossEntropyLoss(weight=class_weights)
```

### Phase 2 – Full Fine-Tuning:

```
optimizer = torch.optim.AdamW(model.parameters(), lr=1e-4)
scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(
    optimizer, T_max=15)

# Best-weight checkpointing
if val_acc > best_val_acc:
    best_val_acc = val_acc
    best_model_wts = copy.deepcopy(model.state_dict())
```

### Inference Pipeline:

```
img = Image.open(img_path).convert("RGB")
input_tensor = preprocess(img).unsqueeze(0).cuda()

with torch.no_grad():
    probs = torch.softmax(model(input_tensor), dim=1)
```

```
conf, pred = torch.max(probs, 1)

print(f"Predicted : {classes[pred.item()]}")
print(f"Confidence: {conf.item()*100:.2f}%")
```

## Social / Industry Relevance of the Project

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### Social Impact

Skin cancer is the most prevalent cancer in many countries. Melanoma, if detected late, has a five-year survival rate below 25%, yet if caught early, survival exceeds 98%. An automated, accurate dermoscopic classifier addresses two major barriers to early detection: geographic access (rural and low-income regions lack specialist dermatologists) and throughput (screening at population scale is infeasible manually). This project demonstrates a pathway to a mobile-deployable AI screening tool that can triage high-risk lesions for specialist review, potentially saving lives through early intervention.

### Industry Relevance

The skills and architectural patterns developed in this project are directly transferable to high-demand industry roles:

- **Medical AI & Digital Health:** Companies such as Google DeepMind, PathAI, and Zebra Medical Vision deploy Vision Transformer architectures for radiology, pathology, and dermatology screening — the exact workflow implemented here.
- **Computer Vision Engineering:** The two-phase fine-tuning strategy, class-imbalance handling, and DataLoader construction are standard production practices in any vision-centric ML team.
- **MLOps & Model Deployment:** Model persistence (.pth), inference pipeline design, and ONNX-ready architecture preparation are core skills for ML engineers at product companies.
- **Research & Academia:** The reproducible benchmark evaluation (classification report, confusion matrix) on HAM10000 aligns with ISIC challenge methodology and is directly publishable.

### HAM10000 — The Seven Lesion Classes

| Code  | Full Name                                     | Malignancy    |
|-------|-----------------------------------------------|---------------|
| akiec | Actinic Keratoses / Intraepithelial Carcinoma | Pre-malignant |
| bcc   | Basal Cell Carcinoma                          | Malignant     |
| bk1   | Benign Keratosis-like Lesions                 | Benign        |
| df    | Dermatofibroma                                | Benign        |
| mel   | Melanoma                                      | Malignant     |
| nv    | Melanocytic Nevi                              | Benign        |
| vasc  | Vascular Lesions                              | Benign        |

## Learning and Reflection

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### Technical Learnings (Engineering & Architecture)

Building this pipeline from scratch required moving beyond surface-level deep learning tutorials and engaging with real production-engineering challenges. Key technical takeaways include:

- **Swin Transformer Internals:** I gained a deep understanding of how Shifted Window attention replaces standard global self-attention to achieve linear complexity with image size, making it feasible to process high-resolution dermoscopic images without memory overflow.
- **Two-Phase Fine-Tuning:** Learning the rationale behind freezing the backbone first—protecting pretrained representations while establishing a well-calibrated head—was a major conceptual breakthrough that directly mirrors techniques used in large language model fine-tuning.
- **Class Imbalance Engineering:** I learned how to mathematically derive inverse-frequency weights from the training distribution and inject them into PyTorch's `CrossEntropyLoss`, a technique that is standard practice in any imbalanced medical dataset pipeline.
- **PyTorch DataLoader Optimisation:** Configuring `num_workers` and `pin_memory` correctly for GPU-pinned asynchronous data loading significantly reduced training time per epoch.
- **Model Checkpointing:** I learned the importance of saving best-validation weights rather than end-of-training weights, and how `copy.deepcopy` prevents aliasing bugs when storing intermediate model states.

### Management and Product Learnings

Developing this pipeline also provided significant insights into ML project management and scope control:

- **Ruthless Prioritisation:** The initial temptation was to implement Grad-CAM explainability, ONNX export, and multi-dataset training simultaneously. Focusing on delivering a correct, reproducible core pipeline first—then extending—was vital to shipping a working prototype within the project timeline.
- **Decoupled Pipeline Design:** By cleanly separating data preparation, model training, evaluation, and inference into distinct notebook cells, any stage can be debugged or re-run independently without restarting the entire training loop.
- **Designing for Clinical Trust:** Reporting per-class precision, recall, and F1—not just top-1 accuracy—taught me that in medical AI, aggregate metrics are insufficient. A model that achieves 90% accuracy by ignoring rare malignant classes is clinically dangerous.

## Overall Experience Summary

My overall experience building this Swin Transformer skin lesion classifier was deeply rewarding. It bridged the gap between being a standard deep learning practitioner and thinking like a Medical AI Engineer.

The most fulfilling moment was watching the Phase 2 validation accuracy steadily climb above the Phase 1 head-only baseline—concrete proof that gradually unlocking the pre-trained backbone extracts progressively more domain-specific features. This project solidified my interest in Medical Computer Vision and gave me the confidence to tackle imbalanced dataset engineering, Vision Transformer fine-tuning, and end-to-end clinical AI deployment in future roles.

## Conclusion and Future Scope

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### Conclusion: Objectives and Achievements Recap

This project successfully delivered a full end-to-end skin lesion classification pipeline on the HAM10000 benchmark using the Swin Transformer architecture. The primary objective of building a reproducible, clinically meaningful classifier was achieved through:

- **Transfer Learning:** Leveraging ImageNet-22k pretrained Swin Transformer weights as a powerful feature extractor, adapted to dermoscopy via two-phase fine-tuning.
- **Imbalance Mitigation:** Dynamically computed inverse-frequency class weights successfully prevented the model from collapsing to majority-class predictions.
- **Rigorous Evaluation:** Per-class precision, recall, and F1 scores alongside a confusion matrix provided a clinically meaningful performance profile beyond top-1 accuracy.
- **Production-Ready Inference:** A standalone prediction cell accepts any dermoscopic image and returns the predicted lesion class with full softmax confidence scores for all seven categories.

This project stands as a robust proof-of-concept for AI-assisted dermatology screening, demonstrating that state-of-the-art Vision Transformer models can be adapted to specialised medical imaging tasks within a well-structured, cost-effective training pipeline.

### Future Scope of the Project

While the current architecture provides a highly functional baseline, the project has significant potential for future scaling and feature expansion:

- **Grad-CAM / Attention Rollout Explainability:** Integrate gradient-based saliency maps to highlight the specific lesion pixels driving each prediction, enabling clinicians to verify model reasoning and increasing diagnostic trust.
- **ONNX / TorchScript Export:** Export the trained model to ONNX format and serve it via a lightweight FastAPI inference endpoint or a browser-based ONNX Runtime, eliminating the need for a GPU server in production.
- **Expanded Dataset Integration:** Incorporate ISIC 2019 and ISIC 2020 challenge datasets alongside HAM10000, and apply MixUp or CutMix augmentation to further reduce class imbalance and improve minority-class recall.
- **Multi-Label Classification:** Extend the model to handle lesions with co-occurring diagnoses using a sigmoid multi-label head, more accurately reflecting real-world clinical complexity.
- **Federated Learning:** Implement a federated training protocol to allow multiple hospital systems to collaboratively train the model without sharing raw patient images, addressing clinical data privacy regulations.